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TEST REPORT

DETERMINATION OF POINT LOAD STRENGTH INDEX OF ROCK — $I_{s(50)}$

Test Standard: ASTM D5731-16 | STP06-RKT-D5731 rev 2 | **Scope: Diametral & Axial Rock Core Tests Only**

Report No.		**Job Ref. No.**	
Client		**Date of Issue**	
Project			
Site Location		**Contract / W.O. No.**	
Borehole No. & Rock Type		**Date Tested**	
Test Procedure	STP06-RKT-D5731 rev 2 (Core Specimens Only)	**Moisture Condition**	

1. Apparatus Traceability & Mandatory Conical Platen Coaxiality Verification

Tester & Load Cell	LT-PO-02 / LC-01	Due: _____	**60° Platens (LT-PL-01~04) Coaxial Check**	$d_{max} - d_{min}$ = _____ mm (< 0.20 mm)
Digital Caliper	LT-DC-01 / 03	Due: _____	**Stopwatch (LT-TD-01, Fail Time 10–60 s)**	Irregular Lump / Block Excluded

2. Individual Rock Core Specimen Test Data (Diametral: L ≥ 0.5D | Axial: 0.33 ≤ D/W ≤ 1.0)

No.	Core Depth (m)	Mode (D / A)	D (mm)	W (mm)	D' (mm)	L (mm)	Time (s)	Load P (kN)	D_e (mm)	I_s (MPa)	Factor F	$I_{s(50)}$ (MPa)
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												

3. Batch Statistical Summary (STP06 Cl. 6.4.2 — Discard 2 Highest & 2 Lowest Values when n ≥ 10)

Valid Core Specimens (n)		**Excluded Extremes (2 High / 2 Low)**	
Mean $I_{s(50)}$ (2 Sig. Fig., MPa)		**Fracture Loading Direction Note**	☐ Parallel ☐ Perpendicular ☐ Isotropic

Measurement Uncertainty & Remarks: Formulas: Diametral $De^2 = D \cdot D'$; Axial $De^2 = 4 \cdot W \cdot D'/\pi$; Is = P/De²; F = (De/50)^0.45; Is(50) = F·Is (individual to 3 sig. fig.). Uncertainty evaluated per MU-RKT-D5731-1 (k = 2, ~95% confidence).

Tested & Recorded By (Qualified Operator):

Name & Signature Date: _____

Checked & Approved By (HOKLAS Approved Signatory):

Authorized Panel: Lam Hoi Fan, James (JL) / Chu Kar Wai, Peter (PC)

Authorized Signature & Company Chop Date: _____

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